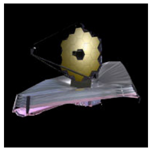

9223 - Let there be Light: Directly Witnessing the Birth of Metal-Free, Pop III Stars in an Ultra-Faint Galaxy at z=6.5

Cycle: 3, Proposal Category: DD

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
Observation Folder				
	1	glimpse_p3d_g395m, glimpse_p2e_g395m, glimpse_p1c_g395m	NIRSpec MultiObject Spectroscopy	(8) abells1063_glimpse_merged_v2

ABSTRACT

Detecting Pop III stars forming out of pristine gas after the Big Bang is a holy grail of astrophysics. Simulations predict these stars should be observable even as late as z~6. However, JWST is yet to push to the low metallicities and masses of the Milky Way's ultra-faint satellites that bear hints of Pop III stars that exploded at z>6. It is only this month, with the deepest imaging taken yet (~31 mag, GO #3293), that JWST has finally breached the ultra-faint regime.

With this Discovery DDT program we propose to confirm the first unambiguous Pop III galaxy at z=6.5. The 9-band NIRCam imaging (including 2 medium-bands) shows the textbook signature of a Pop III cluster: strong Hydrogen emission (H-alpha, EW~2500 A) along with apparently non-existent metal-lines ([OIII]5008, EW~0 A), accompanied by a Balmer jump marking a nascent (<5 Myr) stellar population. No galaxy with similar colors exists in the entire archive -- it is only JWST's deepest imaging to date, amplified 3x by lensing, that revealed this 1e+5 M_sun candidate. With NIRSpec M-grating observations we will rule out [OIII] flux and leave no doubt about its Pop III nature. Alternative hypotheses are

revolutionary in their own right: a new metallicity record, or the discovery of an Intermediate Mass Black Hole.

This program will lay the ground for a new era of Pop III discoveries -- it will validate our novel selection method using strategically chosen medium-bands, and establish the low-mass dwarf regime as the target for Pop III surveys. With profound implications for virtually every subfield of astrophysics, this program promises to bring us closer to the first stars than we have ever been before.

OBSERVING DESCRIPTION

This proposal aims to confirm strong H-alpha and H-beta emission and rule out any significant [OIII]5008 emission to verify the Pop III nature of the target. We require deep sensitivity from 3.5 to 5.0 microns, making NIRSpec with the G395M grating ideal. G395M's resolution allows us to distinguish weak OIII, unlike prism, and achieves approximately twice better S/N for H-beta, enabling stringent constraints on OIII/Hbeta and gas-phase metallicity. This setup also supports our secondary objective to test for an intermediate-mass black hole (IMBH) through a broad Halpha line.

We request 30 hours of exposure (38.7 hours including overhead) to detect H-beta at S/N of 10 and set an upper limit of OIII/Hbeta = 0.1, corresponding to 12 + log(O/H) < 5.6 (or [Z/Zsun] <-3.1). This has a significant chance of falling into the theoretical PopIII territory for the first time, enabled by recent deep NIRCam data, and is an impactful initial test for this Director's Discretionary Time program.

MSA slits will include galaxies with zph > 4 as fillers to explore the unique dwarf galaxy regime in this lensing field. We use a 3-point nod pattern and 494 groups per integration with the NIRSIRS2 readout. The target field is available until November 24, 2024, with the next observing window in May 2025; the current window is preferred, but May is also acceptable.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(8)	abells1063_glimpse_merged_v2	RA: 22 48 54.9024 (342.2287600d) Dec: -44 32 36.66 (-44.54352d) Equinox: J2000		

Comments:
Description=[]

Observation

Proposal 9223, Observation 1: glimpse_p3d_g395m, glimpse_p2e_g395m, glimpse_p1c_g395m Fri May 23 22:00:41 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec MultiObject Spectroscopy

Diagnostics

(glimpse_p3d_g395m, glimpse_p2e_g395m, glimpse_p1c_g395m (Obs 1)) Warning (Form): Config c1 : glimpse_p2e_g395m (#1) has 2 primary slit traces affected by failed open shutters.

(glimpse_p3d_g395m, glimpse_p2e_g395m, glimpse_p1c_g395m (Obs 1)) Warning (Form): Config c1 : glimpse_p2e_g395m (#2) has 2 primary slit traces affected by failed open shutters.

(glimpse_p3d_g395m, glimpse_p2e_g395m, glimpse_p1c_g395m (Obs 1)) Warning (Form): Config c1 : glimpse_p2e_g395m (#3) has 2 primary slit traces affected by failed open shutters.

(glimpse_p3d_g395m, glimpse_p2e_g395m, glimpse_p1c_g395m (Obs 1)) Warning (Form): Config c1 : glimpse_p2e_g395m (#4) has 2 primary slit traces affected by failed open shutters.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(8)	abells1063_glimpse_merged_v2	RA: 22 48 54.9024 (342.2287600d) Dec: -44 32 36.66 (-44.54352d) Equinox: J2000		

Comments:
Description=[]

Acquisition

#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	Filter: F110W; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F110W	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	
2	Filter: F110W; Readout: NRSRAPIDD6; 8 sources in 3 quads; [Optimal TA Accuracy]	SAME	F110W	Auto Acq MSA Config	NRSRAPIDD6	3	1	4	687.153	

Template

TA Method	HFF Readout Mode	Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List	Spectral Overlap Map	Spectral Overlap Threshold
MSATA	false	No	MSA Center	abells1063_glimpse_merged_v2 (1256 sources)		jwst-nirspec-mr	1.5

Reference Stars

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
1	11637	342.251343	-44.550903	22.54810848136924	1	67431	342.234894	-44.527821	22.62481117248535
1	12411	342.183075	-44.549644	23.15032386779785	1	203111	342.240004	-44.552606	23.138671875
1	30914	342.185547	-44.533215	22.51900863647461	1	204623	342.172745	-44.545470	22.272567749023438
1	42939	342.228516	-44.538204	22.514860153198242	1	205320	342.172156	-44.543142	22.57051883988693

Visit	ID	RA	Dec	Magnitude	Visit	ID	RA	Dec	Magnitude
2	5280	342.214630	-44.562370	23.480148315429688	2	26915	342.176575	-44.532440	22.78251946112156
2	12411	342.183075	-44.549644	23.15032386779785	2	30914	342.185547	-44.533215	22.51900863647461
2	14644	342.252502	-44.546917	23.698997497558594	2	33363	342.180176	-44.534279	23.065015524556394
2	17084	342.249939	-44.544693	23.805732437336275	2	210342	342.183067	-44.533113	21.48016357421875

Spectral Elements

#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
1	1 (G395M/F290LP)	c1 : glimpse_p2e_g395m	3 Shutter Slitlet	342.22096095833336 Degrees - 44.54897694444446 Degrees	48.98956922031089			3	6	8403.201
2	1 (G395M/F290LP)	c1 : glimpse_p2e_g395m	3 Shutter Slitlet	342.22096095833336 Degrees - 44.54897694444446 Degrees	48.98956922031089			3	6	8403.201
3	1 (G395M/F290LP)	c1 : glimpse_p2e_g395m	3 Shutter Slitlet	342.22096095833336 Degrees - 44.54897694444446 Degrees	48.98956922031089			3	6	8403.201
4	1 (G395M/F290LP)	c1 : glimpse_p2e_g395m	3 Shutter Slitlet	342.22096095833336 Degrees - 44.54897694444446 Degrees	48.98956922031089			3	6	8403.201
5	1 (G395M/F290LP)	c1 : glimpse_p3d_g395m	3 Shutter Slitlet	342.226232 Degrees - 44.549240277777756 Degrees	48.98587135523372			3	6	8403.201
6	1 (G395M/F290LP)	c1 : glimpse_p3d_g395m	3 Shutter Slitlet	342.226232 Degrees - 44.549240277777756 Degrees	48.98587135523372			3	6	8403.201
7	1 (G395M/F290LP)	c1 : glimpse_p3d_g395m	3 Shutter Slitlet	342.226232 Degrees - 44.549240277777756 Degrees	48.98587135523372			3	6	8403.201
8	1 (G395M/F290LP)	c1 : glimpse_p3d_g395m	3 Shutter Slitlet	342.226232 Degrees - 44.549240277777756 Degrees	48.98587135523372			3	6	8403.201
9	1 (G395M/F290LP)	c1 : glimpse_p1c_g395m	3 Shutter Slitlet	342.2172537916667 Degrees - 44.54572333333334 Degrees	48.9921763289636			3	6	8403.201
10	1 (G395M/F290LP)	c1 : glimpse_p1c_g395m	3 Shutter Slitlet	342.2172537916667 Degrees - 44.54572333333334 Degrees	48.9921763289636			3	6	8403.201
11	1 (G395M/F290LP)	c1 : glimpse_p1c_g395m	3 Shutter Slitlet	342.2172537916667 Degrees - 44.54572333333334 Degrees	48.9921763289636			3	6	8403.201
12	1 (G395M/F290LP)	c1 : glimpse_p1c_g395m	3 Shutter Slitlet	342.2172537916667 Degrees - 44.54572333333334 Degrees	48.9921763289636			3	6	8403.201
13	2 (G395M/F290LP)	c1 : glimpse_p1c_g395m	3 Shutter Slitlet	342.2172537916667 Degrees - 44.54572333333334 Degrees	48.9921763289636			3	6	7527.867

Special Requirements	
	Group Visits within 53.0 Days Visits Same PA MSA Scheduled Aperture PA 48.9841 to 48.9841 Degrees (V3 270.40955 to 270.40955)